Galactic Redshift

Exam code: 0625 & 0972

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Galactic redshift

- When a stationary object emits waves, they spread out **symmetrically**
- If the source of the waves moves, the waves can become squashed together or spread out
- This change in wavelength is known as the **Doppler effect**
- This effect applies to **both** sound waves and electromagnetic radiation (i.e. light waves)

Doppler effect of light

- If a source of light moves **towards** an observer, the observed wavelength **decreases**
  - The wavelength of light **shifts** towards the **blue** end of the spectrum
  - This is called **blueshift**
- If a source of light moves **away** from an observer, the observed wavelength **increases**
  - The wavelength of light **shifts** towards the **red** end of the spectrum
  - This is called **redshift**

Redshift and blueshift of light

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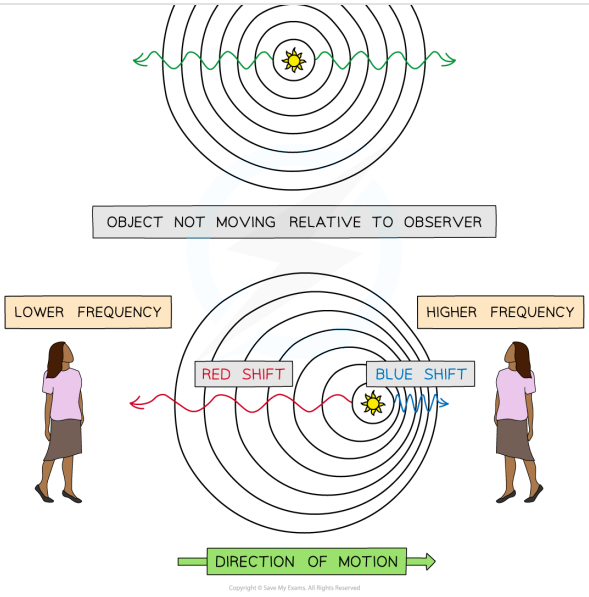
Ask questions and get help with exactly what you're revising.
Got it

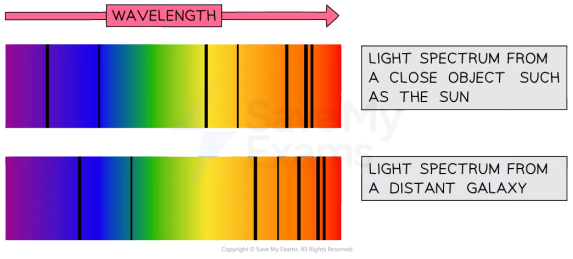


When a light source moves away from an observer, redshift is observed, and when a light source moves towards an observer, blueshift is observed

Observing redshift in distant galaxies

- When astronomers compare light from glowing hydrogen in **distant galaxies** with light from glowing hydrogen on **Earth**, the light appears to be **redshifted**
- This means the observed wavelength has **increased** as the light travelled from the galaxy to the Earth
- This shows that distant stars and galaxies are **moving away** (receding) from the Earth

Redshift of light from a distant galaxy



Light emitted from glowing hydrogen in distant objects appears to be shifted towards the red end of the spectrum, showing they are moving away from Earth

- The greater the observed redshift:
  - the **greater** the **distance** to a galaxy
  - the **faster** the galaxy is moving **away** from Earth

has the **longest wavelength** and the **smallest frequency**.

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